Jaz Winn Ng

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PROFILE

Computer Science undergraduate specialising in low-level systems programming, high-performance computing, and C++. Experienced in collaborating in cross-functional teams, demonstrating problem-solving skills and solving complex engineering challenges under tight deadlines. Passionate about deep learning infrastructure, computer vision, and low-level system optimisation.

EDUCATION

Singapore Institute of Technology Singapore (DigiPen) August 2023 – April 2027

- Bachelor of Computer Science in Real-Time Interactive Simulation
- **Relevant Modules:** Machine Learning | Parallel Programming | Low-Level Programming

SKILLS

- **Programming Language:** C/C++, Python, TypeScript
- **Concepts:** GPU Architecture, Memory Optimisation, Parallel Computing, Advanced Data Structures
- **Frameworks & Libraries:** CUDA, PyTorch, TensorRT, ONNX Runtime, Hugging Face, OpenCV, OpenGL
- **Tools:** Git, Nsight, GDB, Valgrind, CMake, Docker, CI/CD

WORK EXPERIENCE

C++ Developer Intern, Quantified Energy Labs Pte Ltd | Singapore April 2026 – Present

- Migrated drone application from Windows to Linux ARM, reducing payload heat generated and improving power efficiency by 15-20%.
- Built a multi-stage frame alignment, sampling, and stitching video pipeline with coarse-to-fine refinement passes to generate high-fidelity composite modules.
- Built an AI-powered module corner detector with robust geometric fitting, reducing manual parameter calibration to streamline end-to-end post-processing automation.
- Integrated an ESP microcontroller with a field power supply to enable remote power control, centralising device management and enabling end-to-end test automation.

PROJECTS

Optimized Transformer Layer

- Engineered a custom Transformer GPU kernel, leveraging tensor cores and on-chip memory to bypass VRAM bottlenecks.
- Developed an autonomous AI-driven optimization loop to systematically test and integrate 25 performance enhancements.
- Achieved a 10.04× average speedup (46.9× peak) across 14 test cases with zero accuracy degradation against the baseline.

Kos-Engine | **Kos-Engine-2.0**

- Architected a high-performance C++ engine from scratch, focusing on cache-friendly memory layouts, custom allocation strategies, and compute efficiency.
- Engineered a data-oriented Entity-Component-System (ECS) utilising a contiguous SparseSet structure, maximising CPU cache line utilisation and reducing component lookup overhead.
- Profiled and optimised runtime performance by identifying and resolving bottlenecks, while integrating hot-reloading logic to accelerate iterative development cycles.
- Directed project milestones and Git workflows, authored comprehensive Technical Design Documents (TDDs), and established structured code review standards to streamline cross-team collaboration.